



A Two-Stage Deep Learning Framework for Urban Water Pipeline Leak Localization and Multi-Class Diagnosis

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ABSTRACT

Urban water pipeline leaks cause severe resource waste and economic losses annually. Current detection relies on manual inspection using listening rods to capture acoustic signals, but it cannot reliably identify leak conditions and is ineffective for deeply buried pipelines. Meanwhile, cross-correlation-based localization is highly sensitive to environmental noise, often yielding inaccurate results. In this work, we propose a two-stage framework for pipeline leak localization and multi-class leak classification that integrates a controlled experimental platform with a deep learning-based acoustic perception model. The leak localization stage leverages only leak signals, where a Two-Stage Pipeline Conformer (TSPC) learns leak-specific representations from inter-channel time-delay features for localization. The subsequent leak condition diagnosis stage performs multi-class classification using single-channel Mel-spectrograms fused with representations from localization. To evaluate the approach under realistic conditions, a 60 m closed-loop pipeline platform was constructed, and acoustic data were collected under various leak conditions—including normal (no-leak) scenarios and leaks with different orifice sizes—yielding approximately 10 hours of leak signals and 6 hours of normal data. The results show that our model accurately identifies leaks and distinguishes between orifice types, achieving a localization mean absolute error of 1.4 m.

Keywords: Pipeline leak detection, Deep learning, Conformer, Leak localization, Condition diagnosis, Acoustic perception.

1. INTRODUCTION

1.1. Background

Urban water pipeline leaks cause annual global economic losses exceeding 14 billion USD, yet the reliable condition diagnosis and precise localization of leaks in deeply buried pipelines remain long-standing unresolved challenges [1, 2]. Therefore, early detection, precise localization, and condition diagnosis of leaks are crucial. Recently, with the rise of the Internet of Things (IoT) and smart cities, intelligent water management based on sensor networks has become a research hotspot [3, 4]. Among various non-destructive testing technologies (such as ground penetrating radar and distributed

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fiber optic sensing [5]), acoustic methods are widely applied due to their cost-effectiveness and broad applicability [6]. Leak-induced acoustic waves carry rich physical information regarding the leak location and orifice severity [7]. Deeply analyzing these acoustic signals is the core of intelligent pipeline monitoring.

1.2. Limitations of Existing Methods

Current leak detection methods face three major limitations. First, **manual inspection** relies heavily on human experience. Because acoustic signals propagating in buried pipelines suffer significant attenuation due to coupling with the surrounding medium [8], faint leak sounds are easily masked in deeply buried pipelines or noisy traffic environments, leading to a high missed detection rate.

Second, **traditional cross-correlation methods (e.g., GCC-PHAT [9])** face a dual dilemma in practice: strong background noise and multipath reverberation severely blur the correlation peaks; moreover, these methods heavily rely on accurate prior knowledge of wave speed. However, wave speed is highly variable due to pipe material and dynamic water pressure [10], and even a minute estimation error can lead to severe localization deviation.

Third, **existing deep learning methods** generally treat localization and diagnosis as isolated tasks. Previous data-driven approaches have explored Support Vector Machines (SVM) [11], 1D-CNNs [12], and Long Short-Term Memory (LSTM) networks [13] for leak detection. Recently, some studies have also attempted to introduce Vision Transformers (ViT) to process acoustic spectrograms [14]. However, these fragmented paradigms ignore a crucial physical connection: the spatial wave propagation laws and time-delay priors learned by the model during the localization task are highly valuable for guiding the spectral feature extraction needed for orifice diagnosis [15, 16].

1.3. Proposed Framework and Contributions

To overcome the aforementioned limitations, we propose an acoustics-aware Two-Stage Pipeline Conformer (TSPC) framework [17]. The framework is comprehensively validated on a self-built 60 m closed-loop pipeline platform under normal and four different leak orifice conditions. The main contributions of this paper are threefold:

- **A two-stage joint monitoring framework:** Breaking the isolated paradigm, Stage 1 performs high-precision distance regression localization, while Stage 2 conducts multi-class orifice diagnosis, providing an integrated solution for pipeline health assessment.
- **A physics-driven three-channel input strategy:** By fusing the left waveform, right waveform, and cross-correlation sequence, the network implicitly learns wave propagation characteristics. This achieves robust, noise-resistant localization without requiring wave speed priors, yielding a mean absolute error (MAE) of 1.4 m.
- **A cross-task weight transfer mechanism:** By transferring pre-trained weights from the localization task to the diagnosis task, spatial wave propagation laws are injected as physical constraints into frequency-domain feature extraction. This significantly accelerates model convergence and boosts multi-class accuracy.

2. EXPERIMENTAL SETUP AND DATASET

To comprehensively verify the effectiveness of the proposed two-stage framework, this study built a 60-meter closed-loop water supply pipeline experimental platform and collected acoustic data under three pipeline conditions. The resulting data were organized differently for the two stages: Stage 1 used the main leak dataset with an internal train/validation/test split, whereas Stage 2 used all channel-expanded samples from the main dataset as training data and relied on additionally acquired validation and test subsets collected at previously unseen measurement positions to strictly prevent potential data leakage.

2.1. Closed-loop Pipeline Platform

The main topological structure of this experimental platform is a 60-meter-long circulating galvanized steel pipeline with an outer diameter of 76 mm and an inner diameter of 65 mm. To simulate the complex and dynamic fluid environment in urban pipe networks, the system water pressure was irregularly and randomly adjusted between 0.15 and 0.3 MPa during the experiment.



Figure 1: Experimental setup and hardware configuration. The top two images show the overall view of the 60 m closed-loop pipeline. The bottom three images, from left to right, show the sensor installation positions, the piezoelectric acoustic sensor model, and the data acquisition card model.

Regarding sensor deployment, considering that the localization task in Stage 1 must rely on the spatial time-delay difference on both sides of the leak point, we installed acoustic sensors on both sides of the leak, forming a dual-channel monitoring array. The Channel 1 sensor was placed on the left side of the leak, and the Channel 2 sensor was placed on the right side. Therefore, the raw data collected by the system in each recording is a dual-channel tensor containing synchronous signals from both sides. In terms of hardware configuration, acoustic vibration signals were picked up by customized piezoelectric acoustic sensors (model: [WTSV112E]), which were then connected to a high-precision data acquisition card (model: [USB-4431]). The system sampling rate was strictly set to 25600 Hz, and continuous recording was performed using Signalpad software.

2.2. Leak Simulation Mechanism

To obtain high-fidelity acoustic samples under different physical conditions, this study designed a detachable replacement module mechanism in the middle section of the loop pipeline. We set up flange disconnection points at specific locations on the pipeline to facilitate the flexible installation of different test pipe sections.

To correspond to the three-class intelligent diagnosis task in Stage 2, this experiment designed a simulation mechanism for three pipeline conditions: normal, slit-type leak, and crack-type leak. Specifically, we customized and machined four types of replacement modules with specific physical features. In addition to the intact pipe section used to collect background noise and normal water-flow sounds, the **slit-type leak** condition was simulated using a longitudinal slit on the pipe wall, which physically mimics the scenario of a longitudinal seam failure or stress-induced axial cracking



Figure 2: Leak simulation modules, illustrating one normal intact pipe and four different types of physical leak orifices. For the three-class diagnosis task, the longitudinal slit is used to simulate the slit-type leak condition, while the Y-shaped and T-shaped cracks are grouped as crack-type leaks.

commonly observed in welded steel pipelines. The **crack-type leak** condition was simulated using Y-shaped and T-shaped cracks on the walls of the other pipe sections to realistically simulate irregular brittle fractures or fatigue cracks caused by corrosion or external mechanical damage in municipal pipe networks. Previous studies in fluid dynamics and acoustic emission have demonstrated that the geometric shape of the orifice significantly alters the turbulent boundary layer of the jet, thereby exciting acoustic waves with distinct frequency-band energy distributions [18]. These modules were replaced sequentially according to experimental needs. This rigorous experimental design allows us to precisely control the orifice morphology as the sole variable, ensuring that the acoustic feature differences captured by the sensors originate primarily from different physical leak states.

2.3. Dataset Construction

Based on the above platform, continuous acoustic data were collected under the three designed conditions. For Stage 2 diagnosis, representative frequency-domain distributions of the three classes are shown in Figure 3. Compared with the normal state, different leak morphologies exhibit distinct spectral energy distributions and band-dependent intensity patterns. This observation supports the use of Mel-spectrograms as the input representation for the diagnosis network.

It should be emphasized that the dataset organization is different for Stage 1 and Stage 2.

For **Stage 1 leak localization**, the main dataset consists of synchronized dual-channel recordings collected under leak conditions. This dataset was split into training, validation, and test subsets in an 8:1:1 ratio for regression model development and evaluation.

For **Stage 2 leak diagnosis**, special care was taken to avoid hidden data leakage. Specifically, all dual-channel recordings from Stage 1 were first unfolded into independent single-channel samples and used *only as the training set* for Stage 2. In other words, the Stage 2 training data originate entirely from the previously collected Stage 1 recordings after channel-wise expansion. The Stage 2 validation set and test set were *not* obtained by further splitting these recordings. Instead, they were collected additionally and independently. Moreover, the sensor measurement positions used in the Stage 2 validation and test sets were deliberately chosen such that they had never appeared in the Stage 2 training set. This protocol ensures that the diagnosis model is evaluated on genuinely

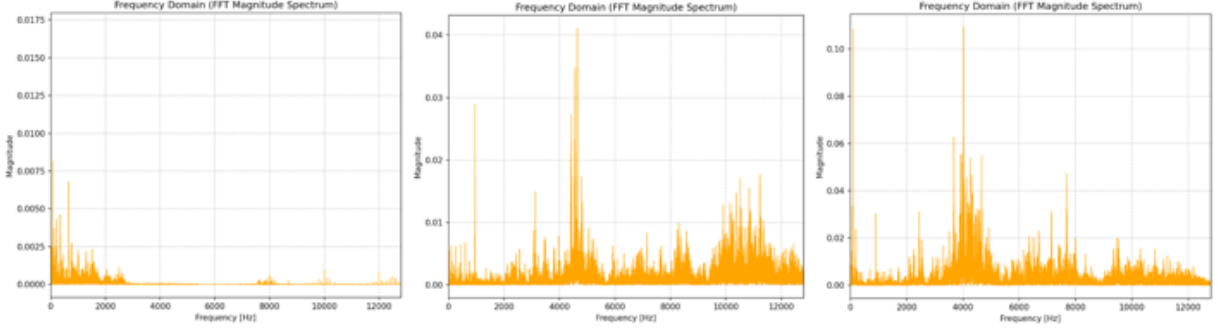


Figure 3: Representative frequency-domain examples of the three pipeline conditions used in Stage 2 diagnosis: normal, slit-type leak, and crack-type leak.

unseen spatial configurations and prevents performance inflation caused by overlapping measurement locations.

To quantitatively summarize the data usage in the two stages, Table 1 reports the dataset construction strategy. The complete dataset contains three classes: normal, slit-type leak, and crack-type leak. For Stage 1, the main dataset was partitioned into training, validation, and test subsets. For Stage 2, all available single-channel samples expanded from the main dataset were used for training, while dedicated validation and test subsets were separately collected, each containing 0.5 h of audio per class. For subsequent engineering analysis, all signals were segmented into 2 s, 5 s, and 10 s frames.

Table 1: Dataset organization for Stage 1 localization and Stage 2 diagnosis. The dataset is categorized into three conditions. The main dataset was split for Stage 1, while expanded single-channel samples were used for Stage 2 training. Validation and test sets were additionally collected at unseen positions.

Label	Condition	Main Dataset (h)	Stage 2 Validation (h)	Stage 2 Test (h)
0	Normal	6.0	0.5	0.5
1	Slit-type	6.0	0.5	0.5
2	Crack-type	6.0	0.5	0.5
Total		18.0	1.5	1.5

3. PROPOSED TWO-STAGE FRAMEWORK

This section details the proposed TSPC framework for pipeline leak localization and condition diagnosis. To break the limitation of traditional methods that separate localization and classification, we design a highly cohesive two-step training architecture with a rigorous data splitting strategy. The overall experimental workflow is illustrated in Figure 4.

3.1. Overall Two-Stage Framework

The core philosophy of our framework is to utilize the spatial wave propagation laws learned during the localization task to guide the subsequent condition diagnosis. The framework consists of two sequential training steps with distinct data utilization strategies:

- **Step 1 (Stage 1):** Focuses exclusively on the distance regression task. It utilizes *only the leak signals* from the original dual-channel dataset, which are partitioned into training, validation,

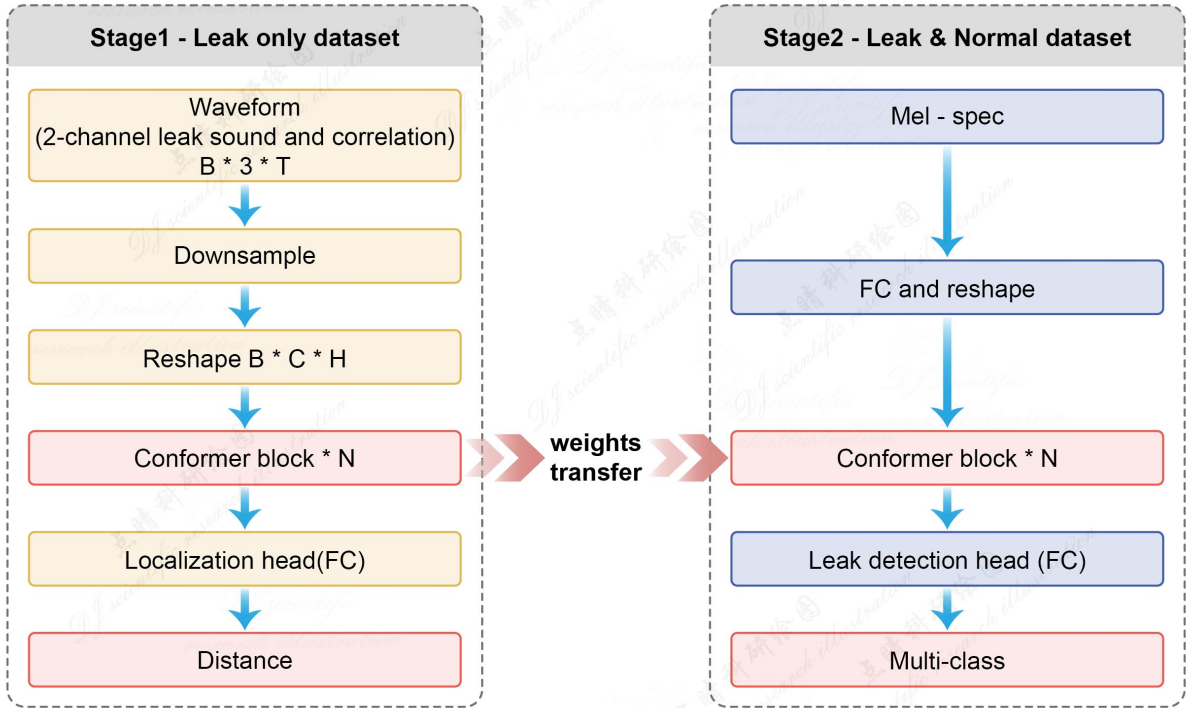


Figure 4: Overall experimental workflow and architecture of the proposed two-stage framework. In Stage 1, the original dual-channel leak audio is split into training, validation, and test sets. The left and right channels are concatenated with their cross-correlation sequence to form a three-channel input, which passes through a downsampling module and Conformer blocks to predict the absolute distance to the left sensor. For Stage 2, all original dual-channel data (including both normal and leak conditions) are separated into single channels to form the Stage 2 training set exclusively. To prevent data leakage, additionally collected single-channel data with unseen sensor placements serve as the validation and test sets. These single-channel signals are converted into Mel-spectrograms and fed into the Conformer backbone (initialized with transferred weights from Stage 1), followed by a classification head for multi-class diagnosis.

and test sets. The input consists of the left channel waveform, the right channel waveform, and their cross-correlation sequence, concatenated into a three-channel tensor. This tensor is processed through a shared backbone (comprising a downsampling module and Conformer Blocks) and terminates with a regression prediction head.

- **Step 2 (Stage 2):** Shifts to the multi-class condition diagnosis task (one normal state and four leak states) using single-channel acoustic data. To strictly prevent data leakage and evaluate true spatial generalization, the dataset is uniquely constructed: *all* original dual-channel recordings (both normal and leak) are separated into independent single channels and used *exclusively as the Stage 2 training set*. Meanwhile, the Stage 2 validation and test sets consist of additionally collected single-channel data where the sensors were placed at previously unseen locations. The network converts these signals into Mel-spectrograms, reuses the backbone weights pre-trained in Stage 1, and terminates with a classification head.

3.2. Stage 1: Leak Localization via Multi-Channel Input

In the localization task of Stage 1, the model aims to predict the absolute distance from the leak point to the left sensor. The input strictly consists of paired dual-channel acoustic waveforms synchronously collected by the left and right sensors under leak conditions. To eliminate amplitude magnitude differences caused by varying water pressures, Z-score normalization is first applied to the raw waveforms:

$$x_{norm} = \frac{x - \mu}{\sigma} , \quad (1)$$

where μ is the mean and σ is the standard deviation of the signal. Since traditional cross-correlation methods are susceptible to noise, we compute the discrete cross-correlation sequence for the normalized left and right waveforms as a physical prior:

$$R_{xy}[m] = \sum_{n=-\infty}^{\infty} x[n]y[n+m] , \quad (2)$$

The left waveform, right waveform, and cross-correlation sequence are then concatenated along the channel dimension to form a three-channel input tensor.

As for the network flow, this tensor first passes through a convolutional downsampling module to compress the sequence length and map features to a high-dimensional space. The downsampled features are then fed into the core Conformer Blocks to extract deep spatial time-delay representations. Finally, the feature sequence undergoes Global Average Pooling and is fed into a regression prediction head (MLP) to output the predicted distance. This stage is optimized using the Mean Absolute Error (MAE) loss:

$$MAE = \frac{1}{N_{batch}} \sum_{i=1}^{N_{batch}} |y_i - \hat{y}_i| , \quad (3)$$

where N_{batch} is the batch size, y_i is the true distance, and $\hat{y}_i$ is the predicted distance.

3.3. Stage 2: Leak Condition Diagnosis via Weight Transfer

In Stage 2, the task transitions to a three-class classification problem. To adapt to single-point monitoring scenarios in practical engineering, the model processes independent single-channel data. As emphasized in the overall framework, to rigorously evaluate the model's generalization ability, all channel-expanded samples from the original dataset were strictly used as the training set. Dedicated validation and test sets were additionally collected at novel, unseen spatial coordinates to prevent any hidden data leakage caused by overlapping sensor positions.

Unlike Stage 1, the subtle morphological differences of leak orifices are more prominent in the frequency domain. Therefore, we extract Mel-spectrograms for the single-channel audio using the Librosa library. Mel-spectrograms have been widely proven to be one of the most effective

representations for capturing faint texture features in non-stationary environmental sounds and industrial acoustic signals [19].

Regarding the weight transfer mechanism, random initialization easily falls into local optima due to the weak acoustic feature differences among different orifices. To overcome this, we directly load the Conformer backbone weights pre-trained in Stage 1 into Stage 2 as initialization weights. Subsequently, a new classification head is attached to the end of the network. During fine-tuning, the network utilizes the spatial wave propagation laws learned previously to focus faster on key frequency bands. This stage is optimized using the multi-class Cross-Entropy (CE) loss function:

$$L_{CE} = -\frac{1}{N_{batch}} \sum_{i=1}^{N_{batch}} \sum_{c=1}^M y_{i,c} \log(p_{i,c}) , \quad (4)$$

where M is the number of classes (three in this study), $y_{i,c}$ is the binary indicator (1 if sample i belongs to class c , 0 otherwise), and $p_{i,c}$ is the predicted probability that sample i belongs to class c .

4. RESULTS AND DISCUSSION

4.1. Implementation Details and Evaluation Metrics

All models in this study were implemented using the PyTorch framework on a server equipped with an NVIDIA GeForce RTX 3090 GPU. The training phase utilized the AdamW optimizer with an initial learning rate set to 0.0001 and a cosine annealing strategy. The batch size was set to 8, and the maximum number of training epochs was 16. After exhaustive grid search, all core comparative experiments in this paper were conducted under the optimal hyperparameter configuration: an input signal frame length of 5 seconds and a network depth of 4 Conformer Blocks. Additionally, in Stage 2, the Mel-spectrograms were generated using a Fast Fourier Transform (FFT) window size (n_{fft}) of 1024 and a hop length of 256 to ensure high-resolution time-frequency representations.

To comprehensively evaluate model performance, the distance regression localization task in Stage 1 uses Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) as evaluation metrics. The condition diagnosis task in Stage 2 involves the potential risk of class imbalance; therefore, in addition to Accuracy, we introduced the Macro-F1 score and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) to more rigorously assess the model's comprehensive generalization ability in multi-class tasks.

4.2. Stage 1: Localization Performance and Baseline Comparison

To verify the superiority of the proposed model in the localization task, we compared our proposed TSPC model with the traditional physical method (GCC-PHAT) and classic deep learning regression models (1D-CNN and ResNet-18).

Table 2: Stage 1: Localization performance comparison with baselines.

Method	MAE (m)	RMSE (m)
GCC-PHAT (Traditional)	3.3	3.6
1D-CNN	2.4	2.7
ResNet-18	2.8	3.0
TSPC (Ours)	1.4	1.8

As shown in Table 2, the traditional GCC-PHAT method performed the worst, with an MAE as high as 3.3 m. This is because, in a real closed-loop pipe network, strong background noise (e.g., pump noise) and multipath reverberation severely distort the peaks of the cross-correlation function. Moreover, this method relies heavily on the pipeline wave speed, which is extremely

difficult to measure accurately under dynamic water pressure. In contrast, data-driven deep learning models exhibited better noise resistance. However, conventional 1D-CNN and ResNet struggle to fully capture global time-delay dependencies in long sequences due to their limited receptive fields, resulting in MAEs of 2.4 m and 2.8 m, respectively. Relying on the Conformer architecture’s powerful ability to model both global and local features, our proposed TSPC implicitly learned the acoustic wave propagation laws in the pipe network, achieving an extremely low MAE of 1.4 m and demonstrating outstanding engineering robustness.

4.3. Stage 2: Diagnosis Performance and Baseline Comparison

In the condition diagnosis task, we evaluated the model’s performance on both the binary classification (Normal vs. Leak) and the more challenging three-class classification (Normal, Square-type, and Crack-type) tasks. We comprehensively compared TSPC with mainstream audio/time-series classification networks (1D-CNN and ResNet-18).

Table 3: Stage 2: Diagnosis performance comparison of different models.

Task	Model	Accuracy (%)	Macro-F1 (%)	ROC-AUC
Binary (2-class)	1D-CNN	96.8	95.4	0.952
	ResNet-18	98.9	98.2	0.983
	TSPC	98.6	98.3	0.97
Multi (3-class)	1D-CNN	91.5	90.4	0.9591
	ResNet-18	94.3	92.2	0.9632
	TSPC	97.3	97.0	0.988

As shown in Table 3, for the binary classification task, all deep learning models achieved high and comparable performance, confirming that the distinction between normal and leak states is relatively straightforward in the acoustic feature space. Therefore, we focus the following analysis on the more challenging three-class condition diagnosis task.

In the three-class task, the acoustic feature differences caused by different orifice morphologies are more subtle. Limited by local convolutional kernels, 1D-CNN and ResNet-18 failed to fully capture full-band acoustic texture details, yielding accuracies of only 91.5% and 94.3%, respectively. In contrast, our TSPC model (combined with Stage 1 transfer weights) achieved a remarkable accuracy of 97.3%, with a Macro-F1 score of 97.0% and a ROC-AUC of 0.988. The consistently high Macro-F1 score demonstrates that the transferred spatial priors effectively guide the extraction of subtle frequency-domain features, allowing the model to robustly distinguish between normal, square-type, and crack-type conditions without suffering from severe class imbalance or inter-class confusion.

To further analyze the class-wise diagnosis performance, Figure 5 presents the confusion matrix of the proposed TSPC model on the three-class test set. The model achieves perfect recognition of the normal state, with all 180 normal samples correctly classified. For the two leak categories, only minor confusion is observed between square-type and crack-type leaks. Specifically, 175 out of 180 square-type samples and 112 out of 120 crack-type samples are correctly identified. The corresponding class-wise F1-scores are 100.0% for normal, 96.4% for square-type, and 94.5% for crack-type, with an overall Macro-F1 of 97.0%. These results indicate that the proposed transfer-enhanced Conformer backbone can effectively capture subtle spectral differences among different leak morphologies while maintaining strong robustness across all classes.

4.4. Comprehensive Ablation Study

To deeply investigate the impacts of core components and hyperparameters on the two-stage framework, we conducted separate ablation studies for Stage 1 localization and Stage 2 diagnosis.

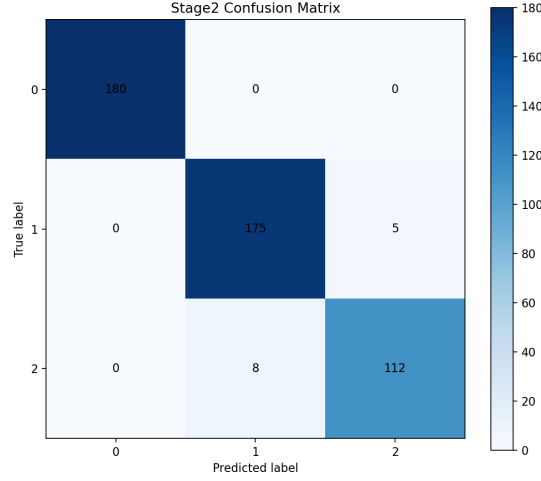


Figure 5: Confusion matrix of the proposed TSPC model on the three-class test set. The model achieves an overall accuracy of 97.3% and a Macro-F1 score of 97.0%, with perfect recognition of the normal class and only minor confusion between square-type and crack-type leaks.

Table 4 summarizes the effects of input features, network depth, and frame length on localization performance, while Table 5 reports the influence of transfer strategy, network depth, and frame length on multi-class diagnosis performance.

4.4.1. Ablation Study on Stage 1 Localization

Table 4: Ablation study for Stage 1 localization performance.

Ablation Target	Configuration	MAE (m)	RMSE (m)
A. Input Features	2-Channel (L+R)	3.5	3.7
	4-Channel (+ Auto-corr)	3.2	3.4
	3-Channel (TSPC Default)	1.4	1.8
B. Network Depth	2 Conformer Blocks	3.1	3.4
	6 Conformer Blocks	3.3	3.5
	4 Conformer Blocks	1.4	1.8
C. Frame Length	2 seconds	2.6	3.0
	10 seconds	1.7	2.2
	5 seconds	1.4	1.8

*N/A indicates failure to converge due to vanishing gradients.

Table 4 presents the ablation results for Stage 1 localization.

Input Features: The comparative experiment revealed a highly important and counter-intuitive phenomenon: the 4-channel model containing more features (adding single-channel auto-correlation to the 3-channel basis) actually produced a higher localization error than the 3-channel model. This result profoundly reveals the core physical essence of the acoustic localization task: spatial time-delay is the only core physical quantity. Auto-correlation features mainly reflect the signal reverberation characteristics at a single sensor location, which acts as interference noise for calculating spatial distance and distracts the self-attention mechanism. Forcing the model to focus only on the 3-channel input essentially injects the purest spatial physical prior into the network.

Network Depth: Regarding network depth, we fixed the frame length at 5 s and tested configurations with 2, 4, and 6 Blocks. Experiments revealed that with 2 Blocks, the model exhibited underfitting with a higher MAE (3.1 m), indicating that a shallow network lacks sufficient receptive field to capture complex spatial wave propagation patterns. When the number of Blocks increased to 6, the MAE increased to 3.3 m, which was worse than the 4-Block configuration. This suggests that excessive depth does not necessarily yield further improvement on this engineering-scale dataset. The 4-Block configuration therefore achieved the best trade-off between representation capacity and trainability.

Frame Length: Regarding signal length, we fixed the network at 4 Blocks and compared inputs of 2 s, 5 s, and 10 s. Results showed that a short frame length of 2 s failed to encompass sufficiently complete low-frequency acoustic wave cycles, leading to unstable cross-correlation feature extraction and increased localization error. Meanwhile, although a 10 s frame length increased the information volume, it also introduced excessive transient environmental noise and redundant background sounds, lowering the signal-to-noise ratio. Therefore, 5 s represents the optimal balance between information completeness and noise resistance.

4.4.2. Ablation Study on Stage 2 Diagnosis

Table 5: Ablation study for Stage 2 multi-class diagnosis performance.

Ablation Target	Configuration	Accuracy (%)	Macro-F1 (%)	AUC
A. Transfer Strategy	Random Initialization	92.7	90.5	0.934
	Frozen Backbone	85.3	82.7	0.959
	Fine-tuning (TSPC Default)	97.3	97.0	0.988
B. Network Depth	2 Conformer Blocks	94.2	90.2	0.948
	6 Conformer Blocks	93.4	91.2	0.952
	4 Conformer Blocks	97.3	97.0	0.988
C. Frame Length	2 seconds	89.5	88.9	0.935
	10 seconds	95.2	94.8	0.976
	5 seconds	97.3	97.0	0.988

Table 5 presents the ablation results for Stage 2 multi-class diagnosis.

Transfer Strategy: In the transfer learning ablation experiment, when the Stage 1 weights were completely frozen and only the classification head was trained, the accuracy plummeted to 82.2%. This is physically reasonable because Stage 1 emphasizes low-frequency time-delay phase differences, whereas Stage 2 requires high-frequency local energy distribution patterns. The feature spaces are therefore not perfectly aligned and must be fine-tuned. Although random initialization can also converge to a relatively high accuracy (92.5%), the transfer fine-tuning strategy (97.4%) still performs best. The Stage 1 weights provide the model with a foundational understanding of spatial wave propagation laws, offering a more favorable initialization space for diagnosis.

Network Depth: Similar to Stage 1, insufficient depth leads to limited feature modeling ability, whereas excessive depth causes optimization degradation. The 4-Block setting again achieved the best diagnosis performance, confirming that moderate model depth is most suitable for this engineering-scale dataset.

Frame Length: For diagnosis, the 5 s frame length also delivered the best performance. Shorter segments may lose discriminative spectral details related to subtle orifice morphology differences, while longer segments tend to introduce more irrelevant environmental interference. Thus, 5 s again

provides the best balance between discriminability and robustness.

5. CONCLUSIONS

This paper proposes an acoustics-aware Two-Stage Pipeline Conformer (TSPC) framework, successfully achieving high-precision localization and multi-class condition diagnosis of leaks in urban water distribution networks. Through a physics-driven three-channel fusion input strategy, the model implicitly learns spatial time-delay priors and effectively overcomes the vulnerability of traditional cross-correlation methods to noise interference without requiring known wave speeds, achieving outstanding localization robustness. Furthermore, this paper innovatively designs a cross-task feature reuse mechanism based on weight transfer, injecting the spatial wave propagation laws learned in the localization stage as physical constraints into the diagnosis task. This significantly accelerates model convergence and substantially improves the accuracy of multi-class diagnosis. In-depth hyperparameter analysis further demonstrates that a signal frame length of 5 seconds and a network depth of 4 Conformer Blocks is the optimal configuration balancing information completeness, noise resistance, and computational efficiency, exhibiting extremely high practical engineering application value. In the future, we will further expand the dataset and verify and optimize this framework in more complex real-world urban underground pipe network topologies (for instance, integrating Graph Neural Networks to process spatial topologies [20]), aiming to achieve large-scale intelligent monitoring of pipe networks.

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